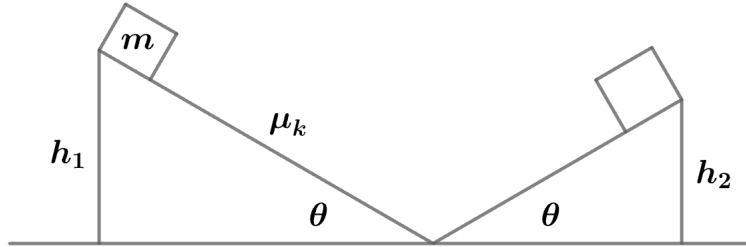


2011 F=ma Exam: Problem 18

Kevin S. Huang



The loss in gravitational potential energy is due to the energy dissipated by friction. Thus, we have

$$mg(h_1 - h_2) = E_d = fd = (\mu_k mg \cos \theta) \left(\frac{h_1 + h_2}{\sin \theta} \right)$$

Simplifying,

$$\begin{aligned} mgh_1 - mgh_2 &= (\mu_k mg \cot \theta) h_1 + (\mu_k mg \cot \theta) h_2 \\ mg(1 + \mu_k \cot \theta) h_2 &= mg(1 - \mu_k \cot \theta) h_1 \\ h_2 &= \frac{1 - \mu_k \cot \theta}{1 + \mu_k \cot \theta} h_1 = \frac{1 - 0.4 \cot(30^\circ)}{1 + 0.4 \cot(30^\circ)} (1.0 \text{ m}) = 0.18 \text{ m} \end{aligned}$$

so the answer is A.